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Key Points:

- Choice of ensemble aggregation measure has important consequences in analyzing global change scenario ensembles
- Benchmarking from a perceived “neutral scenario” can distort the interpretation of the outcomes
- Energy system uncertainty influences global and regional water, energy, and food sector outcomes, each with different characteristics

Supporting Information:

Supporting Information may be found in the online version of this article.

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Discovering the Multisectoral Impacts of Global Energy Sector Outcomes Through Multiple Ensemble Aggregation Measures

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Abstract Understanding complex human-Earth system interactions often involves analyzing large scenario ensembles that encompass a wide range of plausible futures. These ensembles often require aggregation to summarize information based on specific criteria or conditions. However, previous research using global change scenario ensembles has largely overlooked how the choice of aggregation method influences the interpretation of results. To address this gap, we leverage a large ensemble data set designed to capture broad energy system dynamics generated using the Global Change Analysis Model. We first explore how energy-related uncertainties are propagated to both global and regional water-energy-food sectors. We then conduct a rank correlation analysis across seven ensemble aggregation measures and demonstrate the need to consider multiple measures in global change scenarios. Our results suggest that global water and food sector outcomes in the 21st century vary widely depending on different scenario assumptions. The global energy productivity is projected to improve by the end of the century across all scenarios. Moreover, regions facing water scarcity challenges in 2100 do not always overlap with those facing extreme energy and food sector outcomes. Although rank correlations across seven aggregation measures are relatively stable across sectors, we identify cases where relying on a single measure leads to losing critical information in the full ensemble. Reliance on a single aggregation measure can distort the interpretation of global change scenario outcomes. Instead, adopting multiple ensemble aggregation measures provides a more holistic understanding of global change scenario ensembles.

Plain Language Summary Studying how humans and the environment interact requires looking at many future scenarios. Because these scenarios create a lot of data, researchers use different methods to combine and summarize the results. However, past studies have not fully considered how different ways of summarizing the scenarios can affect conclusions. In this study, we use a large data set of possible future scenarios and analyze global and regional water, energy, and food system. We compare seven different ways to summarize the data. Our findings show that while the future of global water and food systems depend on different possible scenarios, the energy productivity is expected to improve by 2100. We also find that areas with water scarcity in the future are not always the same as those with energy and food problems. Though alternative methods for summarizing scenarios often result in similar conclusions, there are also critical differences in some cases which can lead to misinterpreting the scenarios. Using only one combining method can miss important details in the data or result in wrong conclusions. We suggest using multiple summary measures when evaluating future scenarios to gain more reliable insights from the scenarios.

1. Introduction

Global multisector systems are complex, with interregional cross-scale interactions that make their future evolution highly uncertain (Helbing, 2013; Reed et al., 2022; Srikrishnan et al., 2022). To tackle this uncertainty, scientists and modelers often adopt scenarios to evaluate the consequences of alternative futures, in part because scenarios provide self-consistent narratives of systems evolution without requiring explicit likelihoods (Ebi et al., 2014; O'Neill et al., 2020). Two prominent examples of such scenarios are the Representative Concentration Pathways (RCPs) and the Shared Socioeconomic Pathways (SSPs), which have been widely used in the scientific literature and government reports (IPCC, 2023; O'Neill et al., 2014; Van Vuuren et al., 2011). Relying

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on a few shared scenarios enhances scientific intercomparison and communication by unifying assumptions across disciplines and research groups while allowing for structurally diverse modeling capabilities. However, relying on a few shared scenarios also risks missing key drivers of impactful outcomes, which are often difficult to anticipate in complex, multisector systems (Guivarch et al., 2022). For example, there is no guarantee that the canonical global change scenarios (e.g., the RCPs or the SSPs) will capture the drivers of water scarcity in a particular basin or a range of possible energy evolution pathways, even though those outcomes are of most interest (Birnbaum et al., 2022; Dolan et al., 2021; Woodard et al., 2023). In response to this limitation, many have suggested supplementing common scenario frameworks with large ensembles of global change scenarios that broadly survey the uncertainty space, paired with machine learning tools that can identify drivers of critical outcomes (Ambrósio et al., 2024; Lamontagne et al., 2018; McJeon et al., 2011; Rozenberg et al., 2014; Van Vuuren et al., 2019). These approaches, often referred to as exploratory modeling and scenario discovery (Marchau, 2019), hold great promise but also raise new challenges for the global change scenario community, including how one should derive key insights from scenario ensembles that are explicitly non-probabilistic. Several studies have systematically illustrated that seemingly small decisions around ensemble aggregation (i.e., grouping simulation outputs based on a criterion or a condition) can have profound impacts on the analysis results, especially in the context of decision-making under deep uncertainty (DMDU) (Hadjimichael et al., 2020; Herman et al., 2015; McPhail et al., 2018). In this paper, we aim to leverage those insights to evaluate the consequences of using alternative aggregation measures when simulating multisector impacts through a large ensemble of global change scenarios.

In the DMDU literature, multiple ensemble aggregation methods are more frequently highlighted as “robustness measures” (Bonham et al., 2022, 2024; Borgomeo et al., 2018; Herman et al., 2015; McPhail et al., 2018). Broadly, the most common robustness measures can be classified into expected value (Wald, 1949), satisficing-based (Simon, 1956), and regret measures (Savage, 1951). In satisficing, one seeks outcomes that satisfactorily meet some criteria rather than optimal outcomes. Consequently, satisficing-based robustness measures compute either the range of scenarios that achieve acceptable performance based on predefined criteria (i.e., the domain criterion) (Starr, 1963) or the maximum extent of uncertainty a system can tolerate before failing to meet the criteria (i.e., the Info-Gap) (Ben-Haim, 2004). They are distinct from regret measures, which quantify the cost of choosing incorrectly in the face of uncertainty (Herman et al., 2015). Regret measures compute the difference between the performance from one decision alternative to the most plausible or the best-case alternative's performance. Because each robustness measure is designed to capture different aspects of an ensemble, scientists and modelers can draw different insights depending on the measure (McPhail et al., 2018). Studies comparing different robustness measures and highlighting the consequences of choosing have been articulated in the risk analysis, decision-making, and water resources management literature.

For instance, Herman et al. (2015) evaluated the performance of water resources systems with four variants of regret and satisficing-based robustness measures and highlighted that the methodological choices on robustness measure may lead to different policy suggestions to stakeholders. McPhail et al. (2018) conducted a taxonomy analysis on diverse robustness measures, highlighting the consequences of choosing different definitions of robustness when evaluating the alternatives in multiple water systems-related case studies. Hadjimichael et al. (2020) explored how the factors leading to outcomes of interest can vary according to different definitions of robustness measures. However, we identify an important research gap for the multisectoral, global change scenarios community, which is to better understand the potential consequences of the choice of ensemble aggregation measure. Due to the recent surge in experiments involving large ensembles of global change scenarios (Birnbaum et al., 2022; Dolan et al., 2021, 2022; Guivarch et al., 2022; Lamontagne et al., 2018; Niazi et al., 2024; Van Vuuren et al., 2019; Wessel, Iyer, et al., 2024; Woodard et al., 2023), we believe our work is particularly timely to ensure that key insights are not unduly influenced by the choice of aggregation measure. While no decision is explicitly made using global change scenarios, the insights drawn from the scenarios can ultimately inform a broad range of research activities. Therefore, the robustness of the insights can matter for a broader range of scientific bodies. To this end, we adapt the key concepts of robustness measures from the DMDU literature and use them to derive robust insights from global change scenario ensemble outcomes. Through our experiment, we also demonstrate how the applications of the key concepts from the DMDU literature can extend beyond decision-focused applications to global-scale, multisector analyses.

In our experiment, we leverage a data set that covers a wide range of uncertainty from the energy sector makeup simulated using a leading global multisector model, the Global Change Analysis Model (GCAM) (Calvin

et al., 2019). GCAM has been widely adopted to capture the complex interactions in the human-Earth system under multiple drivers (e.g., Earth system and environmental factors, human influence, and policies) (Weyant, 2017). We utilize an existing data set that samples drivers of the energy sector Wessel, Iyer, et al. (2024) and track changes in the water-energy-food sectors over the 21st century, both globally and regionally. We then perform rank correlation analysis of seven ensemble aggregation measures and emphasize the importance of multi-measure comparison in global change scenarios through illustrative examples. Through this experiment, we contribute to the existing literature by (a) discovering how energy-related factors affect global and regional water-energy-food sectors and (b) highlighting the necessity of adopting multiple measures in global change scenarios ensembles to prevent the choices on measures leading to biased conclusions.

The rest of this paper is organized as follows. In Section 2, we first introduce the simulation model and the data set we utilize during the analysis. We then explain each sector's outcome metric used to track sectoral performance, ensemble aggregation measures, and the rank correlation method that we adopt in our study. In Section 3, we demonstrate how each sector's global metric changes over the 21st century from a multisectoral perspective. We then highlight basins and regions with extreme sectoral outcomes in 2100. We also perform a rank correlation analysis using seven ensemble aggregation measures and highlight instances where relying on a single measure can be misleading. Finally, we summarize our significant findings and suggest potential topics for future studies in Section 4.

2. Methods

This experiment uses a scenario ensemble from the GCAM model (Section 2.1) that evaluates multiple energy-sector uncertainties (Section 2.2). Using this ensemble, we evaluate multi-sectoral impacts measured with a variety of sectoral outcome metrics (Section 2.3), both globally and regionally. We then examine the implications of using a variety of ensemble aggregation measures (Section 2.4) in global change scenarios through rank correlation analysis (Section 2.5) across basins and regions.

2.1. GCAM

GCAM is a global-scale multisector model that simulates highly interconnected human and natural systems to explore the complex interaction of the energy, water, land use, climate, and socioeconomics systems (Calvin et al., 2019). GCAM's dynamic-recursive market-clearing structure is solved in 5-year time steps from the base year of 2015 through 2100, with market shares of competing technologies, energy sources, crop types, and numerous other commodities determined by a logit-choice algorithm incorporating resource supply and cost curves. Our version of the GCAM model divides the world into 33 geopolitical regions (32 core regions plus Uruguay separated from the Southern South America region), 235 water basins, and 384 land-use regions. Ultimately, the model yields an equilibrium price for each market in which the supply meets the demand for each commodity in each geopolitical region, water basin, and land-use region. The data set we utilize is simulated with GCAM v6.0 (version-specific GCAM documentation at <https://jgcri.github.io/gcam-doc/v6.0/toc.html>) with some modifications from GCAM-Latin America and the Caribbean, which separates Uruguay as a standalone region (Delgado et al., 2020; Khan et al., 2020; Santos da Silva et al., 2021; Wild et al., 2021). Due to GCAM's wide coverage of sectors and spatiotemporal scales, we provide a brief overview of how GCAM operates in this section. For a detailed discussion of GCAM's structure, the readers are referred to Calvin et al. (2019).

Our work focuses on the global and regional water-energy-food sectors modeled through GCAM. In GCAM, the water-energy-food sectors are interconnected and thus interact comprehensively in response to socioeconomic drivers, technological changes, and policy decisions (Wild et al., 2021). Specifically for the food sector, supply for each food commodity is determined by the combination of land allocation for agriculture, fertilizer use, and water availability in each land-use region (Dolan et al., 2022). On the other hand, the demand for food is more dependent on economic aspects of a region, such as consumer income and food prices (Narayan et al., 2021). For the energy sector, both nonrenewable and renewable energy sources are transformed into diverse energy carriers that are ultimately used by the final energy demand sectors (e.g., residential and commercial buildings, passenger and freight transportation, and manufacturing and non-manufacturing industries) (Kanyako et al., 2024; Wessel, Iyer, et al., 2024). The energy sector is also connected to both the food sector (through land use competition from commercial biomass and fertilizer production) and the water sector (through the use of cooling water for multiple technologies).

Table 1

Summary of Uncertainties Varied During the Scenario Design From Wessel, Iyer, et al. (2024)

Name	Abbreviation	Discrete sampling levels	Short description
Population and GDP	SOC	SSP1, SSP2 (Reference), SSP3, SSP4, SSP5	Population and GDP pathways by region consistent with five SSP formulations
Energy Preferences	EP	Reference, Literature-based Alternatives	Representative scenarios reflecting alternative Nationally Determined Contribution pathways and long-term strategies for each country achieved and maintained
Value of Terrestrial Carbon	VTC	Gradual (Reference), Rapid Valuation of Terrestrial Carbon (only coupled with non-reference EP)	Value of terrestrial carbon passed to the land use system
Institutional Factors	INS	Equal (Reference), Differentiated risk across regions and technologies	Investment risks in clean energy projects regionally and across technologies
Advanced Hydrogen	ADH	Reference, Advanced Hydrogen	Scaling of Hydrogen technology in the energy system
Industry Energy Efficiency	IEE	Reference, Higher efficiency	Aggregate efficiency improvements across industries (cement, chemicals, fertilizer, etc.)
Buildings Energy Efficiency	BEE	Reference, Higher efficiency	Efficiency for heating and cooling, plug load in households, and floor space
Transport Electrification	TRE	Reference, Higher electrification	Share of electric vehicles changing over time
Wind and Solar Capital Costs	VRE	Low, Medium (Reference), High	Overnight cost of capital for wind and solar technologies
Direct Air Capture Cost	DAC	SSP2 consistent (Reference), SSP3 consistent (High)	Cost of capital for Direct Air Capture technology
Temperature-responsive Energy Demand	CID	No impacts (Reference), Impacts (RCP6.0 for reference EP, RCP2.6 for non-reference EP)	Changes in heating and cooling degree days in each region

2.2. Scenarios

We leverage an existing ensemble of 5,760 GCAM scenarios developed by Wessel, Iyer, et al. (2024), which broadly samples energy-economic uncertainties. The 11 uncertain factors sampled for scenario construction include, among others, population and gross domestic product (GDP), energy preferences, energy efficiency improvements, transport electrification, and heterogeneous investment risk; the full ensemble makeup is summarized in Table 1. The non-reference scenario of energy preferences is based on countries' Nationally Determined Contributions and net-zero goals and includes an additional factor that varies the economic value of terrestrial carbon, which can have a substantial impact on the land system when other sectors are constrained. Exploring this emergent pressure on land availability is consistent with past work showing how uncertainty in one sector can increase uncertainty in others, as shown in Kanyako et al. (2024).

2.3. Sectoral Outcome Metrics

For this study, we adopt three metrics to capture the outcomes of global and regional water-energy-food sectors: physical water scarcity, energy intensity, and food budget share. Whereas the physical water scarcity captures the hydrological sustainability of a basin in terms of water availability ratio, the other two metrics evaluate the economic aspects of a region in terms of energy productivity and food accessibility. For consistency, basins, regions, and scenarios with lower sectoral outcome metrics indicate less water-scarce, highly energy-productive, and highly food-accessible outcomes.

Physical water scarcity, adopted from Vorosmarty et al. (2000) and Hoekstra et al. (2012), has been frequently applied in previous GCAM studies to capture a basin's physical water shortage (Birnbaum et al., 2022; Dolan et al., 2021; Hejazi et al., 2014b). We compute physical water scarcity as the ratio of total water withdrawals to total renewable water supply to measure the sustainability of human-driven water consumption. Renewable water supply includes: surface water (runoff), renewable groundwater, reservoir storage, and baseflow minus environmental flows (Birnbaum et al., 2022; S. H. Kim et al., 2016). For total water withdrawals, we include water withdrawals from all six end-use sectors in GCAM: agriculture, electricity generation, industrial manufacturing,

primary energy production, livestock, and municipal (Hejazi et al., 2014b). Ultimately, physical water scarcity for each basin, year, and scenario is calculated by:

$$\text{Physical water scarcity}(b, y, s) = \frac{TW W(b, y, s)}{TWS(b, y, s)} [-] \quad (1)$$

where TWW is the total water withdrawals; TWS is the total renewable water supply; b , y , and s correspond to a given basin, year, and scenario, respectively.

For the energy sector, we compute the energy intensity of the economy in each region by dividing the final energy consumption by the total economic production (i.e., the GDP), after adjusting for inflation using the Consumer Price Index. This concept of energy intensity has also been adopted in previous studies using GCAM (Calvin et al., 2016). We aggregate each region's energy consumption across all final energy demand sectors to compute the regional final energy consumption. Energy intensity is then calculated by dividing final energy consumption by the region's total GDP, as shown in Equation 2:

$$\text{Energy intensity}(r, y, s) = \frac{\sum_{sect} FEC_{sect}(r, y, s)}{GDP(r, y, s)} [kWh/\$2020] \quad (2)$$

where FEC is the final energy consumption; r and $sect$ correspond to a given region and final energy demand sector, respectively.

Through food budget share, we aim to capture each region's food accessibility through the ratio of food expenditure to income. Our food metric is closely related to the metric utilized from Cui et al. (2022), in which the food accessibility is computed by dividing the staple food expenditure by average income. In our analysis, we expand the scope of food expenditure to include all commodities from staple food (e.g., grains and roots and tubers) and nonstaple food (e.g., fruits, vegetables, oils, dairy, meat, fish, and poultry), based on GCAM's classification (Edmonds et al., 2017). Ultimately, food demand for each commodity i is multiplied by the regional prices corresponding to that specific commodity and divided by the GDP of the region to calculate each region's food accessibility Equation 3:

$$\text{Food budget share}(r, y, s) = \frac{\sum_i FD_i(r, y, s) \times CP_i(r, y, s)}{GDP(r, y, s)} \times 100 [\%] \quad (3)$$

where FD is the food demand and CP is the food commodity's regional prices.

Note that the version of GCAM we use provides regional producer prices for each food commodity rather than consumer prices. Consequently, food budget shares from our work can be relatively lower than actual food budget shares due to the discrepancy between consumer and producer prices, especially in higher income regions where the difference between the two is greater due to more complicated food distribution processes.

2.4. Ensemble Aggregation Measures

When dealing with large scenario ensembles, it is often desirable to aggregate or summarize information across ensemble members. For example, one might be interested in comparing the effect of land constraints on energy systems in various regions. To ensure robust insights, one could use a diverse ensemble of scenarios, some aggregation measures to summarize regional performance, and scenario discovery to identify the drivers of discordant scenarios. One challenge is that the likelihood of the ensemble members is often unknown or disputed; therefore, non-probabilistic approaches are desired. Here, we draw inspiration from the DMDU literature, which has thoroughly explored similar measures, but in the context of decision-making (Herman et al., 2015; McPhail et al., 2018). However, the context in global change scenarios is different: often no decision is being explicitly made, and defining satisfactory performance is challenging because of the sectoral, spatial, and temporal scales being considered. Therefore, direct use of concepts such as satisficing and regret is not appropriate. In this section, we describe seven alternative ensemble aggregation measures that can be summarized into three categories:

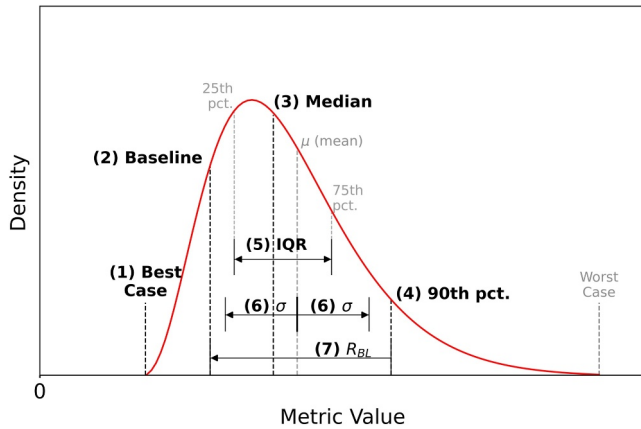


Figure 1. Seven ensemble aggregation measures from three categories used in this study. The best-case, baseline, median, and 90th percentile scenario belong to the relative threshold category; the interquartile range and standard deviation (σ) belong to the dispersion category; R_{BL} belongs to the performance degradation category. For demonstration purposes, we use a positively skewed stylized distribution with the baseline scenario close to the best case to represent the distribution of metrics from an ensemble. Note that not all distributions are positively skewed, and the baseline is not always near the best case.

relative thresholds (analogous to satisficing in DMDU), dispersion, and performance degradation (analogous to regret). We visualize the seven ensemble aggregation measures we adopt in this study in Figure 1.

The first category of aggregation measures we consider are based on relative thresholds, which track the sectoral outcome metrics (i.e., system performance) tied to quantiles of the simulation results or an otherwise notable simulation: (a) best-case, (b) baseline, (c) median, and (d) the 90th percentile. The best-case measure selects the minimum metric value (i.e., system performance) across the scenario ensemble, while the baseline is the metric value tied to our a priori “middle of the road” scenario (the “SSP2” scenario from Calvin et al. (2017)). In the context of global change scenarios, the baseline is often considered a neutral scenario, so tracking its performance across regions and sectors is particularly valuable. The median tracks the central tendency or location of the ensemble members and is a simple measure of which regions or sectors tend to be more impacted by an experimental factor. The final threshold measure we consider is based on the 90th percentile scenario. Assuming larger metric values are undesirable, the upper tail of the distribution is often of interest to scientists and modelers. Any representative large quantile could be selected, but we chose the 90th percentile scenario in this study because it is relatively (but not excessively) extreme. All four relative threshold measures we adopt to aggregate ensemble simulations provide information about the ensemble based on a specific scenario or quantile and are

analogous to satisficing-based robustness measures, which identify outcomes that meet acceptable thresholds rather than optimizing for best outcomes. However, we distinguish them from the satisficing robustness measures in previous literature and adopt a different terminology (i.e., relative thresholds) because no decision is being made with the outcomes from global change scenarios. Consequently, there is typically no satisficing threshold determining whether an outcome is “satisfactory” in a global change scenario ensemble. Instead, we use relative thresholds as scenario quantiles to glean insights from a distribution of outcomes.

The second category of aggregation measures considers the dispersion of the ensemble, or put simply, the extent of disagreement across the ensemble. When evaluating the general dispersion of a distribution, (e) the interquartile range (IQR) and (f) the standard deviation (σ) are often adopted in statistics. Although both measures consider dispersion of the distribution, IQR is less sensitive to outliers that are often present in global change scenario ensembles. In this study, the coefficient of variation and its outlier-resistant alternatives are not considered because they are not suitable for data with a mean near zero, as in this case. Unlike the relative threshold measures, dispersion measures are difficult to interpret in isolation, but they can provide valuable insight when paired with other measures.

The last measure belongs to the performance degradation category and is modified from regret-based robustness measures from Herman et al. (2015). This measure quantifies the performance degradation from a target scenario. From the DMDU literature, “regret” is defined as the relative difference between a specific scenario's performance and the performance of either the most plausible or the best possible scenario (McPhail et al., 2018). Similarly, the performance degradation measure in our study, (g) R_{BL} , is formulated as the degradation of the 90th percentile metric from that of the neutral baseline scenario. R_{BL} captures the degradation (or improvement) from our baseline assumptions to the upper tail of outcomes, and measures how far outcomes may deviate from our a priori best guess. Formulation of our performance degradation measure, R_{BL} , is as follows:

$$R_{BL,i}(br,y) = M_{i,90}(br,y) - M_{i,BL}(br,y) \quad (4)$$

where $i = W, E, F$ correspond to each sector (i.e., water, energy, and food), br is basin or region (depending on the metric), $M_{i,90}$ and $M_{i,BL}$ are the performance metrics from the 90th percentile scenario and the baseline scenario, respectively.

2.5. Rank Correlation

In global change scenario analyses, it is often useful to rank regions, basins, or sectors by the relative magnitude of their outcomes, as this allows scientists and modelers to prioritize their focus in the places that are expected to be most impacted. An important consideration in this study is how the choice of ensemble aggregation measure (Section 2.4) impacts relative ranking, which is particularly important for global change analyses where relative impacts can be more useful than absolute metric values. To do this comparison, we use Spearman's rank correlation (Helsel & Hirsch, 1993) which is computed as the Pearson correlation coefficient of the ranks between two data sets, and thus is bounded $[-1,1]$. Values closer to 1 (-1) indicate a stronger positive (negative) relationship between the two rank orders being compared, while values close to zero suggest no direct relationship. In this analysis specifically, we compare the ranking of basins or regions' outcome metrics (Section 2.3) between pairs of ensemble aggregation measures (Section 2.4).

Similar methodologies have been used in the decision support literature to compare the ranking stability of decision alternatives under different measures of robustness and for multiple objectives, for example, in water resources contexts (Herman et al., 2015; McPhail et al., 2018). However, these analyses typically target specific stakeholder objectives within well-constrained or regulated systems, often resulting in established success benchmarks and widely accepted satisficing thresholds for reliability, resilience, or cost. Because of the exploratory nature of this study and the use of a large, non-probabilistic global change scenario ensemble, we move beyond static thresholds for classifying scenarios and use statistics of the outcome distributions to group scenarios of interest. Rather than using a rank correlation analysis to inform specific, tangible planning alternatives for decision support, we apply this technique in the context of global change scenarios to characterize the uncertainty and potential instability of our measures in identifying the most vulnerable basins or regions.

3. Results and Discussion

We present the results from this study in two sections. Section 3.1 first highlights the temporal dynamics of the water-energy-food sectors across the ensemble globally and reports regions with extreme sectoral outcomes. Section 3.2 tracks the rank correlation of multiple ensemble aggregation measures across regions and motivates the use of alternative measures in global change scenarios.

3.1. Global and Regional Water-energy-Food Sectoral Outcomes

3.1.1. Temporal Dynamics of Global Sectoral Outcomes

Tracking each sectoral outcome metric over time provides insight into how future uncertainties will shape future outcomes. We start by highlighting the global trends in physical water scarcity, energy intensity, and food budget share over the 21st century, across the scenario ensemble and baseline scenario (Figure 2). The global metrics refer to the crop revenue-weighted global physical water scarcity and population-weighted global energy intensity and food budget share (see Text S1 Supporting Information S1 for details on weighting).

From Figure 2, we observe that the variance of all three sectoral outcome metrics generally increases over time and is maximized at the end of the 21st century, but with distinct characteristics. For instance, global physical water scarcity follows a characteristic “peak and decline” behavior (Niazi et al., 2024; Turner et al., 2019) across most scenarios, with overall water scarcity peaking in the 2040s and declining through the later half of the century (Figure 2a). This decline is attributable to the depletion of many large aquifers in the middle of the century, forcing the adoption of more water-efficient technologies and patterns of agricultural production. The scenarios that deviate from this pattern are tied to the socioeconomic assumption with exponential population growth (see Figure S2 in the Supporting Information S1). These scenarios necessitate continued water withdrawals, despite global water depletion and rising costs associated with water consumption. Ultimately, the impacts from energy-economic uncertainties in our ensemble result in high variability in global water scarcity outcomes by the end of the century, which range from 0.40 to 0.60.

On the other hand, we find that the sectoral outcomes for energy and food generally improve over the 21st century. For instance, global energy intensity falls across the entire scenario ensemble (Figure 2b), which largely reflects the assumptions embedded in GCAM about improved techno-economic productivity over time. We find that while all scenarios reflect a marked increase in energy efficiency across the century, the global energy intensity still ranges by a factor of 3 across the ensemble in 2100 ($0.18 \sim 0.54$ kWh/\$). Similarly, the baseline scenario's

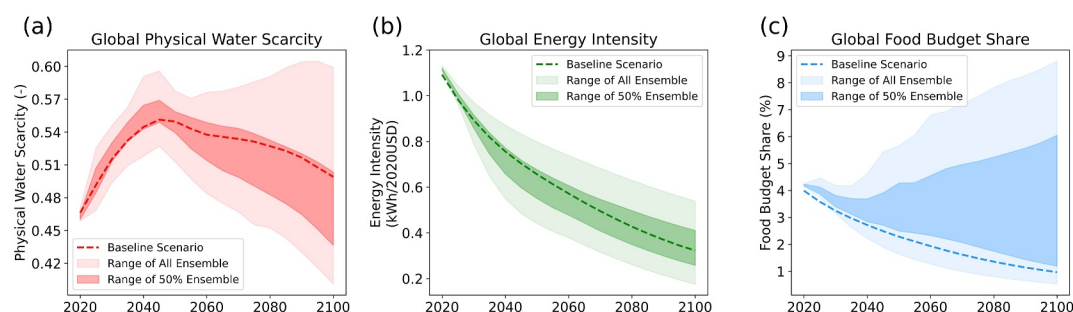


Figure 2. Trajectories of the global sectoral outcome metrics over the 21st century. (a) Physical water scarcity, (b) energy intensity, and (c) food budget share. Outcomes from the baseline scenario are shown as dashed lines. Scenarios with lower metrics indicate more desirable outcomes.

global food budget share also falls from 3.99% in 2020 to 0.97% in 2100, reflecting enhanced food accessibility under the baseline scenario (Figure 2c). However, the food budget share in 2100 varies appreciably across the scenario ensemble, ranging from an increase of 121% to a decrease of 86.7% from the food budget share in 2020. In general, the largest increases in food budget share are tied to increased pressures in the land sector emerging from competition between protected land (e.g., biomass, forest, and grassland) and land used for food production (e.g., cropland and pastureland) (Wise et al., 2014). Scenarios with extensive land competition experience rising food prices worldwide, ultimately affecting the long-term global food budget shares. By highlighting the trajectories of global sectoral outcome metrics evolving over the 21st century, we demonstrate how energy-economic uncertainties affect sectoral outcomes generally. However, scientists and analysts oftentimes focus on extreme outcomes (e.g., scenarios above the 90th percentile) because those outcomes pose significant societal challenges. Therefore, we further explore regions with extreme sectoral outcomes in the following subsection.

3.1.2. Regional Water-energy-Food Sectoral Outcomes in 2100

We shift focus from global to regional sectoral outcomes and identify which parts of the world are relatively more water-scarce, less energy-productive, and less food-accessible by the end of the 21st century. Multiple studies have utilized metric values from a specific percentile (e.g., the median or the 99th percentile) to highlight extreme outcomes from a scenario ensemble (Birnbaum et al., 2022; Hadjimichael et al., 2020). Similarly, Figure 3 reports the 90th percentile for each basin's or region's sectoral outcome metrics in 2100.

From Figure 3a, we first identify basins with high physical water scarcity: the lower Colorado River, Sinai Peninsula, Sabarmati, Africa North Interior, Ziya He Interior, and the Indus. These basins are consistent with past work identifying water scarcity “hotspots” both using GCAM (Hejazi et al., 2014a; S. H. Kim et al., 2016) and outside GCAM (Greve et al., 2018; Huang et al., 2021; Leijnse et al., 2024; Liu et al., 2022). Basins with the most extreme water scarcity outcomes undergo unsustainable water consumption due to rapid population growth and extensive non-renewable groundwater extraction by the end of the 21st century. In general, physical water scarcity outcomes show marked heterogeneity across basins, which reflects that they are influenced by a combination of water supply and regional water demand projections.

Contrarily, extreme outcomes for energy intensity and food budget share are relatively more homogeneous than physical water scarcity outcomes, which in part arises due to the nature of socioeconomic projections across scenarios. In other words, population and GDP projections across regions are more homogeneous than the projection in the water sector: country-to-country variability of population and GDP projections is not reflected in the SSPs as the basin-to-basin variability of water sector projections. We identify Brazil, the Middle East, Russia, and Africa as the regions with the most extreme energy intensity outcomes, largely reflecting relatively lower GDP growth in 2100. Finally, food budget share outcomes are the highest in Southern and Eastern Africa, in which their food budget share surpasses 20% (22.25% and 21.27%, respectively). These results arise from the combination of lackluster GDP growth in certain socioeconomic assumptions, combined with intense land use competition between protected land and cropland, as highlighted in subsection 3.1.1. A more thorough analysis of key drivers in the scenarios exceeding the 90th percentile for each region and sector is described in Supporting Information S1 (Text S2).

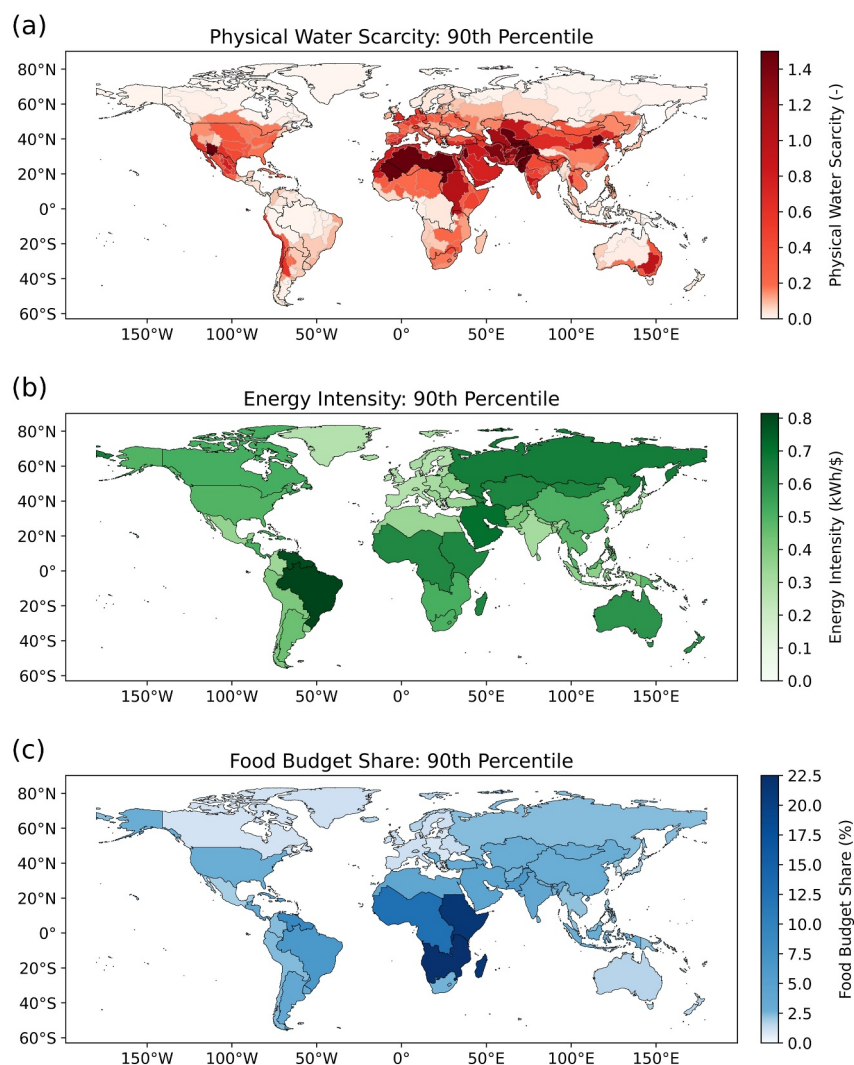


Figure 3. Maps of 90th percentile sectoral outcome metrics across basins and regions in 2100. (a) Physical water scarcity, (b) energy intensity, and (c) food budget share. Note that the spatial resolution of physical water scarcity (235 basin scale) is different from those of energy intensity and food budget share (33 geopolitical region scale). Geopolitical regions in (a) are bounded by black outlines. Basins and regions with darker colors indicate places with less desirable outcomes.

3.2. Implications From Multiple Ensemble Aggregation Measures in Global Change Scenarios

As described in the methods, scientists often wish to summarize an ensemble with some statistic (i.e., ensemble aggregation measure), to report a summary finding or to rank regions by the severity of their outcomes across the ensemble. We utilize the seven aggregation measures introduced in Section 2.4 to aggregate each basin or region's sectoral outcome metrics across the ensemble. We then rank the basins and regions based on each aggregation measure, and compute the correlation of ranks using Spearman's rank correlation coefficient r (Figure 4). The rank correlation results indicate the extent to which rankings based on one aggregation measure remain consistent when using alternative measures.

From Figure 4, we first find that the rank correlations of basins and regions across the same category of aggregation measures are stable across all sectoral outcomes. For instance, the outcome from the baseline scenario and those aggregated through measures from the relative threshold category (i.e., best case, median, and the 90th percentile) agree with strong positive correlations (lower right side of the heatmaps in Figures 4a–4c). This indicates that similar basins or regions are likely to be highlighted regardless of the relative thresholds used during the aggregation. This agreement also holds true for dispersion measures: we observe strong correlations between

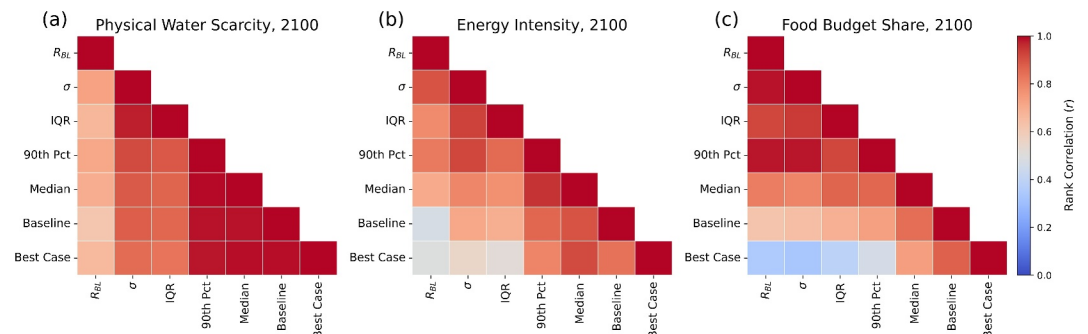


Figure 4. Heatmaps of basins' and regions' rank correlation coefficients across seven ensemble aggregation measures for each sectoral outcome. (a) Physical water scarcity, (b) energy intensity, and (c) food budget share. The rank correlations are computed across 232 basins for physical water scarcity and 33 regions for energy intensity and food budget share. Annotated heatmaps are available in Figure S3 in Supporting Information S1.

the two dispersion measures across all metrics (0.97 for physical water scarcity, 0.93 for energy intensity, and 0.94 for food budget share). Furthermore, the rank correlations gradually decrease as we compare the rankings using measures from different categories (traveling either vertically or horizontally across heatmaps in Figure 4). This strong agreement within the same category and weaker agreement across different categories of aggregation measures is consistent with previous findings on multiple robustness measures in water resources literature (Herman et al., 2015; McPhail et al., 2018). Our results imply that these conclusions can be further expanded to large ensemble simulations with global, multisector models and sectors outside of water.

However, we find that the rank correlations across different categories of measures (e.g., baseline vs. R_{BL} or 90th percentile vs. IQR) yield relatively higher correlations when compared to the results from previous literature (McPhail et al., 2018). This departure from prior findings arises because previous works compute rank correlations of conflicting decision alternatives (e.g., alternatives found by optimizing problems with conflicting objectives). Consequently, the rankings of decision alternatives are less stable when different categories of aggregation measures are used. However, in our application, we compute the rank correlations of basins or regions modeled through a global multisector model instead of conflicting decision alternatives. In the case of global change scenarios, influences from global changes, such as socioeconomics and energy efficiency improvements, have monolithic effects on many basins or regions. In other words, many basins or regions are impacted in the same direction by the influences from external assumptions. This is a reflection of the scenario inputs considered rather than an emergent feature of multisectoral modeling. This explains the higher rank correlations even across different categories of aggregation measures in our application. Despite high rank correlations across measures, we observe cases where rankings differ between aggregation measures. These instances highlight important differences between the measures and the value of a multi-measure analysis of global change scenarios.

Figure 5 reports the relative rankings of all basins and regions across the seven ensemble aggregation measures using a parallel axis plot. Each line in Figure 5 represents a single basin or region and each axis is an aggregation measure. A disagreement between rankings can be identified when lines cross between axes. Overall, Figure 5 confirms the high rank correlations between ensemble aggregation measures across all sectoral outcomes, as previously highlighted through Figure 4. At the same time, we highlight instances with very different rankings across measures, which were overlooked in rank correlation results. For instance, the USA's food budget share ranking is 29th out of 32 under the baseline scenario, but improves to 14th when the standard deviation is used. On the other hand, India's food budget share ranking fluctuates from 6th out of 32 (under the median) to 27th (under the standard deviation) depending on the choice of aggregation measure (Figure 5c). Thus, food budget share in India is relatively low in most scenarios and under the baseline, but is relatively very high in extreme cases and variable overall. The rankings of the USA's food budget share are the opposite, in which the baseline ranking is higher than the ranking from the standard deviation.

Similar ranking disagreements also exist in the water and energy sectors, as illustrated in Figures 5a and 5b. Expanding on this, Figure 6 considers the tradeoff between alternative aggregation measures in the water and energy sector. Figure 6a compares the 90th percentile and the standard deviation of physical water scarcity for different modeled basins, highlighting the Ziya-He and the Indus. The Ziya-He and the Indus basins have nearly

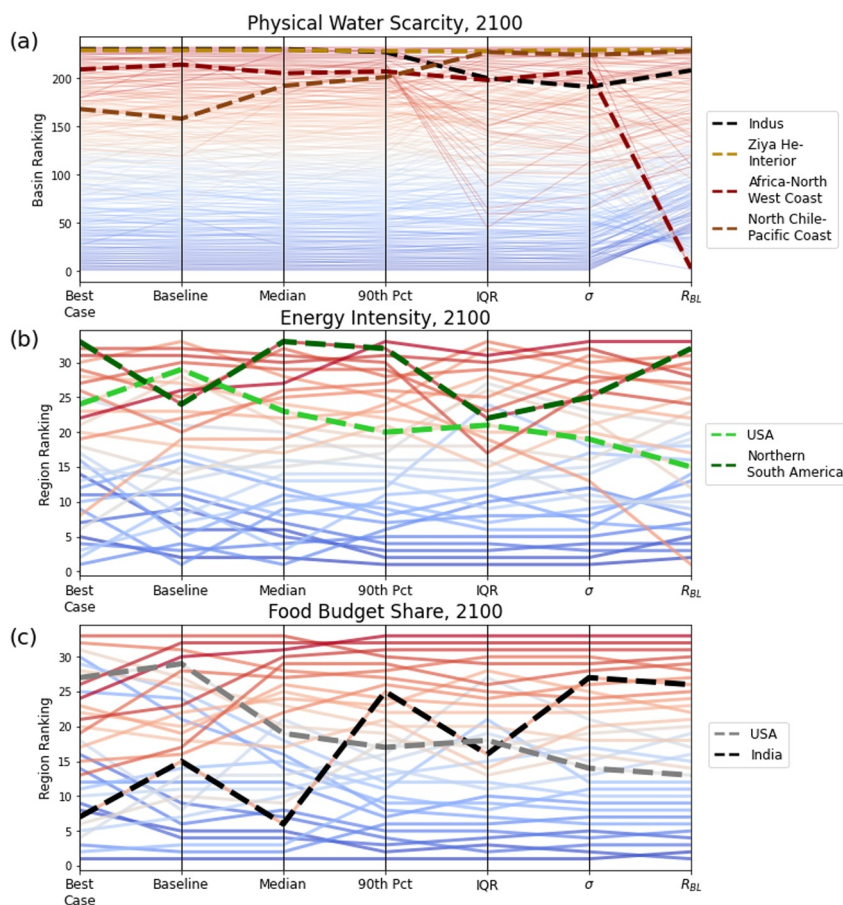


Figure 5. Parallel axis plots showing basin or region ranks computed by seven ensemble aggregation measures. (a) Physical water scarcity, (b) energy intensity, and (c) food budget share. Basins or regions closer to the upper part of the plots (high rankings) indicate basins or regions with more extreme outcomes. Colors represent the ranks from the 90th percentile, in which the region or basin colored in darker red has more extreme outcomes. Thicker, dashed lines depict ranks for selected basins and regions in (a) and (b) are shown in more detail in Figure 6. See Figure S4 in Supporting Information S1 for more examples of basins for physical water scarcity.

the same 90th percentile, but the Ziya-He has much more variability across the ensemble. This indicates that the ensemble simulations disagree more substantially for the Ziya-He, while the simulations for the Indus are extreme but consistent. Such disagreement across ensemble simulations is critical when providing robust analysis or decision support, but it would be lost if only an extreme quantile were considered as a market.

Figure 6b compares the 90th percentile to R_{BL} for physical water scarcity. Recall that in global scenario analyses, the baseline scenario is often the standard scenario assumptions or the “middle of the road” projection. While the Africa-North West Coast and the North Chile-Pacific Coast basins have nearly equal 90th percentiles, their R_{BL} are substantially different, because their baselines are not near the center of the ensemble (Figure 6e). In the case of North Chile-Pacific Coast, the baseline is optimistic compared to the scenario ensemble (i.e., it is less than the ensemble median), while the baseline for the Africa-North West Coast is pessimistic (i.e., it is much greater than the ensemble median). This result highlights that the baseline scenario can sometimes be relatively extreme itself, and should be scrutinized. Figure 6c compares IQR and R_{BL} of regional energy intensity, highlighting the USA and Northern South America. Whereas both regions exhibit similar dispersion when measured by IQR, Northern South America's distribution is asymmetric with a heavy right (upper) tail (Figure 6f). This asymmetry and the projections in the upper tail are valuable in global change scenarios as they reveal the region's worst-case outcomes under the most extreme scenarios. Therefore, capturing this asymmetry through R_{BL} is especially valuable. Overall, our application emphasizes the need to adopt multiple aggregation measures in order to draw holistic insights from large ensemble simulations of global change scenarios.

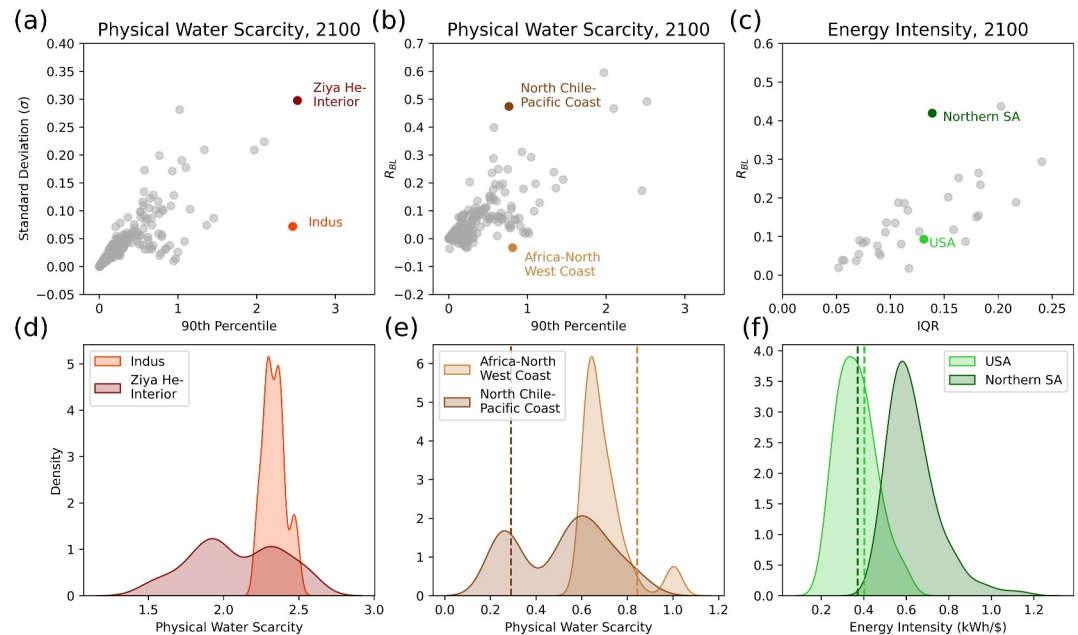


Figure 6. Scatterplots and kernel density estimate plots of basins and regions on multiple ensemble aggregation measures. Scatterplots of (a) 90th percentile (x-axis) versus standard deviation (y-axis) of physical water scarcity, (b) 90th percentile (x-axis) versus R_{BL} (y-axis) of physical water scarcity, (c) interquartile range (x-axis) versus R_{BL} (y-axis) of energy intensity; Kernel density estimate plots of (d) Indus and Ziya He-Interior's physical water scarcity, (e) Africa-North West Coast and North Chile-Pacific Coast's physical water scarcity, and (f) the USA and Northern South America's energy intensity. In (e) and (f), dashed lines correspond to the outcomes from the baseline scenario.

4. Conclusion and Future Work

Understanding the future uncertainties associated with the human-Earth system often involves the use of shared community scenarios (e.g., the RCPs or the SSPs) or large ensembles of global change scenarios. Experiments involving large ensembles of global change scenarios sometimes require aggregation across ensemble members to summarize information across the simulations. Although the methodological choice of ensemble aggregation can have a profound impact on the results, the importance of choosing appropriate methods to aggregate ensemble simulations has been neglected in previous research efforts in the global change scenario community. Because the insights drawn from ensemble simulations often guide a wide range of scientific research activities, understanding how the choice of aggregation measures shapes those insights is important for the broader scientific community.

To address this gap, we utilize a large ensemble simulation from a leading multisector model, the GCAM, that covers wide uncertainties associated with the effects of the energy sector makeup (Wessel, Iyer, et al., 2024). We perform a comprehensive analysis of the global and regional water-energy-food sectors to investigate how they respond to energy system-related uncertainties over the 21st century. We also investigate how the choice of ensemble aggregation measures may affect the conclusions we draw from large ensemble simulations of global change scenarios. We include seven aggregation measures, each with distinct characteristics: four relative threshold measures, two dispersion measures, and a performance degradation measure. We then conduct a rank correlation analysis across the measures and highlight instances where the insights can be misleading when we rely on a single category of aggregation measures.

We first observe that global water availability, energy productivity, and food accessibility exhibit distinctive behaviors in response to energy-related uncertainties. This finding highlights the necessity of designing multi-sectoral modeling experiments to broadly include uncertainties from multiple sectors, as the effects from one sector have diverse effects on various sectors and scales. More specifically, water scarcity is intensified globally until mid-century and improves over the latter half of the 21st century. The world becomes more energy productive despite the challenges introduced by the change of energy preferences. On the other hand, global food accessibility is highly heterogeneous and is heavily impacted by the pressure from the land sector. From a regional perspective, we find that a region's techno-economic productivity and food accessibility are strongly correlated

and are closely tied to the region's economic production. In contrast, water availability in a region is less correlated to the energy and food sectors but is more impacted by nature-driven water supply and human-driven water consumption.

While the overall rank correlations across ensemble aggregation measures are relatively higher than in previous literature, we highlight the consequences of relying on a single category of ensemble aggregation measures. Our analysis demonstrates that important information within the ensemble can be lost when relying on a single aggregation measure, emphasizing the importance of adopting multiple ensemble aggregation measures in global change scenarios. For instance, dispersion measures better capture the disagreement across ensemble simulations compared to relative threshold measures. The performance degradation measure can capture the deviation from the neutral scenario or the asymmetry within the distribution. Through our application, we contribute both a multisectoral analysis identifying the effects of global energy evolutions on the water-energy-food sectors and a methodological proposition regarding the selection of appropriate ensemble aggregation measures in global change scenarios for further analysis.

Through our work, the methodological choice of ensemble aggregation measures in global change scenarios is expected to enlighten future scenario discovery efforts utilizing global-scale, multisectoral models. We believe that performing scenario discovery to identify the key factors leading to extreme outcomes based on different measures (e.g., relative threshold vs. performance degradation) can provide valuable insights when designing or interpreting global change scenarios. Our work has the potential to be further expanded to investigate more detailed spatiotemporal scales in future applications. Although we mostly concentrate on the water-energy-food sectoral outcomes at the end of the 21st century, investing in how the outcomes evolve over the entire simulation horizon of the model through time series clustering tools (Steinmann et al., 2020) holds promise. Moreover, since our main contributions from this work are either at a global or regional level, identifying further implications from sub-regional outcomes is also a promising future topic. We believe our framework of exploring multiple ensemble aggregation measures through a global-scale, multisectoral energy-economic model holds value as this is an active, ongoing research topic to advance local-scale, robustness analyses. (Hadjimichael et al., 2020, 2023).

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

GCAM is an open-source model available at <https://github.com/JGCRI/gcam-core>. The code to run the ensemble and query output databases is available at Wessel, Lamontagne, et al. (2024). The code to process query output data and generate all figures is published on Zenodo at G. J. Kim (2025).

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References

- Ambrósio, G., Doelman, J. C., Schipper, A. M., Stehfest, E., & van Vuuren, D. (2024). Global sustainability scenarios lead to regionally different outcomes for terrestrial biodiversity. *Environmental Research Letters*, 19(10), 104055. <https://doi.org/10.1088/1748-9326/ad73eb>
- Ben-Haim, Y. (2004). Uncertainty, probability and information-gaps. *Reliability Engineering & System Safety*, 85(1–3), 249–266. <https://doi.org/10.1016/j.res.2004.03.015>
- Birnbaum, A., Lamontagne, J., Wild, T., Dolan, F., & Yarlaga, B. (2022). Drivers of future physical water scarcity and its economic impacts in Latin America and the Caribbean. *Earth's Future*, 10(8), e2022EF002764. <https://doi.org/10.1029/2022ef002764>
- Bonham, N., Kasprzyk, J., & Zagana, E. (2022). post-mordm: Mapping policies to synthesize optimization and robustness results for decision-maker compromise. *Environmental Modelling and Software*, 157, 105491. <https://doi.org/10.1016/j.envsoft.2022.105491>
- Bonham, N., Kasprzyk, J., Zagana, E., & Smith, R. (2024). Interactive and multimetric robustness tradeoffs in the Colorado River basin. *Journal of Water Resources Planning and Management*, 150(3), 05023025. <https://doi.org/10.1061/jwrmd5.wreng-6199>
- Borgomeo, E., Mortazavi-Naeini, M., Hall, J. W., & Guillod, B. P. (2018). Risk, robustness and water resources planning under uncertainty. *Earth's Future*, 6(3), 468–487. <https://doi.org/10.1002/2017ef000730>
- Calvin, K., Bond-Lamberty, B., Clarke, L., Edmonds, J., Eom, J., Hartin, C., et al. (2017). The SSP4: A world of deepening inequality. *Global Environmental Change*, 42, 284–296. <https://doi.org/10.1016/j.gloenvcha.2016.06.010>
- Calvin, K., Pachauri, S., De Cian, E., & Mouratiadou, I. (2016). The effect of African growth on future global energy, emissions, and regional development. *Climatic Change*, 136(1), 109–125. <https://doi.org/10.1007/s10584-013-0964-4>
- Calvin, K., Patel, P., Clarke, L., Asrar, G., Bond-Lamberty, B., Cui, R. Y., et al. (2019). GCAM v5. 1: Representing the linkages between energy, water, land, climate, and economic systems. *Geoscientific Model Development*, 12(2), 677–698. <https://doi.org/10.5194/gmd-12-677-2019>
- Cui, R. Y., Waldhoff, S., Clarke, L., Hultman, N., Patwardhan, A., & Gilmore, E. A. (2022). Evaluating the regional risks to food availability and access from land-based climate policies in an integrated assessment model. *Environment Systems and Decisions*, 42(4), 547–555. <https://doi.org/10.1007/s10669-022-09860-4>

- Delgado, R., Wild, T. B., Arguello, R., Clarke, L., & Romero, G. (2020). Options for Colombia's mid-century deep decarbonization strategy. *Energy Strategy Reviews*, 32, 100525. <https://doi.org/10.1016/j.esr.2020.100525>
- Dolan, F., Lamontagne, J., Calvin, K., Snyder, A., Narayan, K. B., Di Vittorio, A. V., & Vernon, C. R. (2022). Modeling the economic and environmental impacts of land scarcity under deep uncertainty. *Earth's Future*, 10(2), e2021EF002466. <https://doi.org/10.1029/2021ef002466>
- Dolan, F., Lamontagne, J., Link, R., Hejazi, M., Reed, P., & Edmonds, J. (2021). Evaluating the economic impact of water scarcity in a changing world. *Nature Communications*, 12(1), 1915. <https://doi.org/10.1038/s41467-021-22194-0>
- Ebi, K. L., Hallegatte, S., Kram, T., Arnell, N. W., Carter, T. R., Edmonds, J., et al. (2014). A new scenario framework for climate change research: Background, process, and future directions. *Climatic Change*, 122(3), 363–372. <https://doi.org/10.1007/s10584-013-0912-3>
- Edmonds, J. A., Link, R., Waldhoff, S. T., & Cui, R. (2017). A global food demand model for the assessment of complex human-earth systems. *Climate Change Economics*, 8(4), 1750012. <https://doi.org/10.1142/s2010007817500129>
- Greve, P., Kahil, T., Mochizuki, J., Schinko, T., Satoh, Y., Burek, P., et al. (2018). Global assessment of water challenges under uncertainty in water scarcity projections. *Nature Sustainability*, 1(9), 486–494. <https://doi.org/10.1038/s41893-018-0134-9>
- Guivarch, C., Le Gallic, T., Bauer, N., Fragkos, P., Huppmann, D., Jaxa-Rozen, M., et al. (2022). Using large ensembles of climate change mitigation scenarios for robust insights. *Nature Climate Change*, 12(5), 428–435. <https://doi.org/10.1038/s41558-022-01349-x>
- Hadjimichael, A., Quinn, J., Wilson, E., Reed, P., Basdekas, L., Yates, D., & Garrison, M. (2020). Defining robustness, vulnerabilities, and consequential scenarios for diverse stakeholder interests in institutionally complex river basins. *Earth's Future*, 8(7), e2020EF001503. <https://doi.org/10.1029/2020ef001503>
- Hadjimichael, A., Reed, P. M., Quinn, J. D., Vernon, C. R., & Thurber, T. (2023). Multi-actor, multi-impact scenario discovery of consequential narrative storylines for human-natural systems planning. *Authorea Preprints*.
- Hejazi, M., Edmonds, J., Clarke, L., Kyle, P., Davies, E., Chaturvedi, V., et al. (2014a). Integrated assessment of global water scarcity over the 21st century under multiple climate change mitigation policies. *Hydrology and Earth System Sciences*, 18(8), 2859–2883. <https://doi.org/10.5194/hess-18-2859-2014>
- Hejazi, M., Edmonds, J., Clarke, L., Kyle, P., Davies, E., Chaturvedi, V., et al. (2014b). Long-term global water projections using six socio-economic scenarios in an integrated assessment modeling framework. *Technological Forecasting and Social Change*, 81, 205–226. <https://doi.org/10.1016/j.techfore.2013.05.006>
- Helbing, D. (2013). Globally networked risks and how to respond. *Nature*, 497(7447), 51–59. <https://doi.org/10.1038/nature12047>
- Helsel, D. R., & Hirsch, R. M. (1993). *Statistical methods in water resources*. Elsevier.
- Herman, J. D., Reed, P. M., Zeff, H. B., & Characklis, G. W. (2015). How should robustness be defined for water systems planning under change? *Journal of Water Resources Planning and Management*, 141(10), 04015012. [https://doi.org/10.1061/\(asce\)jwr.1943-5452.0000509](https://doi.org/10.1061/(asce)jwr.1943-5452.0000509)
- Hoekstra, A. Y., Mekonnen, M. M., Chapagain, A. K., Mathews, R. E., & Richter, B. D. (2012). Global monthly water scarcity: Blue water footprints versus blue water availability. *PLoS One*, 7(2), e32688. <https://doi.org/10.1371/journal.pone.0032688>
- Huang, Z., Yuan, X., & Liu, X. (2021). The key drivers for the changes in global water scarcity: Water withdrawal versus water availability. *Journal of Hydrology*, 601, 126658. <https://doi.org/10.1016/j.jhydrol.2021.126658>
- IPCC. (2023). *AR6 synthesis report (tech. rep.)*. Author. Retrieved 2023-11-10, from Retrieved from <https://www.ipcc.ch/ar6-syr/>
- Kanyako, F., Lamontagne, J., Snyder, A., Morris, J., Iyer, G., Dolan, F., et al. (2024). Compounding uncertainties in economic and population growth increase tail risks for relevant outcomes across sectors. *Earth's Future*, 12(1), e2023EF003930. <https://doi.org/10.1029/2023ef003930>
- Khan, Z., Wild, T. B., Carrazzone, M. E. S., Gaudioso, R., Mascari, M. P., Bianchi, F., et al. (2020). Integrated energy-water-land nexus planning to guide national policy: An example from Uruguay. *Environmental Research Letters*, 15(9), 094014. <https://doi.org/10.1088/1748-9326/ab9389>
- Kim, G. J. (2025). Data and codes for Kim et al. “discovering the multisectoral impacts of the global energy sector outcomes through diverse cross-ensemble aggregation measures” [Dataset]. *Zenodo*. <https://doi.org/10.5281/zenodo.16929421>
- Kim, S. H., Hejazi, M., Liu, L., Calvin, K., Clarke, L., Edmonds, J., et al. (2016). Balancing global water availability and use at basin scale in an integrated assessment model. *Climatic Change*, 136(2), 217–231. <https://doi.org/10.1007/s10584-016-1604-6>
- Lamontagne, J. R., Reed, P. M., Link, R., Calvin, K. V., Clarke, L. E., & Edmonds, J. A. (2018). Large ensemble analytic framework for consequence-driven discovery of climate change scenarios. *Earth's Future*, 6(3), 488–504. <https://doi.org/10.1002/2017ef000701>
- Leijnse, M., Bierkens, M. F., Gommans, K. H., Lin, D., Tait, A., & Wanders, N. (2024). Key drivers and pressures of global water scarcity hotspots. *Environmental Research Letters*, 19(5), 054035. <https://doi.org/10.1088/1748-9326/ad3c54>
- Liu, W., Liu, X., Yang, H., Ciais, P., & Wada, Y. (2022). Global water scarcity assessment incorporating green water in crop production. *Water Resources Research*, 58(1), e2020WR028570. <https://doi.org/10.1029/2020wr028570>
- Marchau, V. (2019). *Decision making under deep uncertainty, from theory to practice*. Springer.
- McJeon, H. C., Clarke, L., Kyle, P., Wise, M., Hackbarth, A., Bryant, B. P., & Lempert, R. J. (2011). Technology interactions among low-carbon energy technologies: What can we learn from a large number of scenarios? *Energy Economics*, 33(4), 619–631. <https://doi.org/10.1016/j.eneco.2010.10.007>
- McPhail, C., Maier, H. R., Kwakkel, J. H., Giuliani, M., Castelletti, A., & Westra, S. (2018). Robustness metrics: How are they calculated, when should they be used and why do they give different results? *Earth's Future*, 6(2), 169–191. <https://doi.org/10.1002/2017ef000649>
- Narayan, K. B., Vernon, C. R., Waldhoff, S. T., Edmonds, J. A., & Cui, R. (2021). Ambrosia: An R package for calculating and analyzing food demand that is responsive to changing incomes and prices. *Journal of Open Source Software*, 6(PNNL-SA-157944), 2890. <https://doi.org/10.21105/joss.02890>
- Niazi, H., Wild, T. B., Turner, S. W., Graham, N. T., Hejazi, M., Msangi, S., et al. (2024). Global peak water limit of future groundwater withdrawals. *Nature Sustainability*, 7(4), 1–10. <https://doi.org/10.1038/s41893-024-01306-w>
- O'Neill, B. C., Carter, T. R., Ebi, K., Harrison, P. A., Kemp-Benedict, E., Kok, K., et al. (2020). Achievements and needs for the climate change scenario framework. *Nature Climate Change*, 10(12), 1074–1084. <https://doi.org/10.1038/s41558-020-00952-0>
- O'Neill, B. C., Kriegler, E., Riahi, K., Ebi, K. L., Hallegatte, S., Carter, T. R., et al. (2014). A new scenario framework for climate change research: The concept of shared socioeconomic pathways. *Climatic Change*, 122(3), 387–400. <https://doi.org/10.1007/s10584-013-0905-2>
- Reed, P. M., Hadjimichael, A., Moss, R. H., Brelsford, C., Burleyson, C. D., Cohen, S., et al. (2022). Multisector dynamics: Advancing the science of complex adaptive human-earth systems. *Earth's Future*, 10(3), e2021EF002621. <https://doi.org/10.1029/2021ef002621>
- Rozenberg, J., Guivarch, C., Lempert, R., & Hallegatte, S. (2014). Building SSPs for climate policy analysis: A scenario elicitation methodology to map the space of possible future challenges to mitigation and adaptation. *Climatic Change*, 122(3), 509–522. <https://doi.org/10.1007/s10584-013-0904-3>
- Santos da Silva, S. R., Hejazi, M. I., Iyer, G., Wild, T. B., Binsted, M., Miralles-Wilhelm, F., et al. (2021). Power sector investment implications of climate impacts on renewable resources in Latin America and the Caribbean. *Nature Communications*, 12(1), 1276. <https://doi.org/10.1038/s41467-021-21502-y>

- Savage, L. J. (1951). The theory of statistical decision. *Journal of the American Statistical Association*, 46(253), 55–67. <https://doi.org/10.1080/01621459.1951.10500768>
- Simon, H. A. (1956). Rational choice and the structure of the environment. *Psychological Review*, 63(2), 129–138. <https://doi.org/10.1037/h0042769>
- Srikrishnan, V., Lafferty, D. C., Wong, T. E., Lamontagne, J. R., Quinn, J. D., Sharma, S., et al. (2022). Uncertainty analysis in multi-sector systems: Considerations for risk analysis, projection, and planning for complex systems. *Earth's Future*, 10(8), e2021EF002644. <https://doi.org/10.1029/2021ef002644>
- Starr, M. K. (1963). Product design and decision theory. (No Title).
- Steinmann, P., Auping, W. L., & Kwakkel, J. H. (2020). Behavior-based scenario discovery using time series clustering. *Technological Forecasting and Social Change*, 156, 120052. <https://doi.org/10.1016/j.techfore.2020.120052>
- Turner, S. W., Hejazi, M., Yonkofski, C., Kim, S. H., & Kyle, P. (2019). Influence of groundwater extraction costs and resource depletion limits on simulated global nonrenewable water withdrawals over the twenty-first century. *Earth's Future*, 7(2), 123–135. <https://doi.org/10.1029/2018ef001105>
- Van Vuuren, D. P., Bijl, D. L., Bogaart, P., Stehfest, E., Biemans, H., Dekker, S. C., et al. (2019). Integrated scenarios to support analysis of the food–energy–water nexus. *Nature Sustainability*, 2(12), 1132–1141. <https://doi.org/10.1038/s41893-019-0418-8>
- Van Vuuren, D. P., Edmonds, J., Kainuma, M., Riahi, K., Thomson, A., Hibbard, K., et al. (2011). The representative concentration pathways: An overview. *Climatic Change*, 109(1–2), 5–31. <https://doi.org/10.1007/s10584-011-0148-z>
- Vorosmarty, C. J., Green, P., Salisbury, J., & Lammers, R. B. (2000). Global water resources: Vulnerability from climate change and population growth. *science*, 289(5477), 284–288. <https://doi.org/10.1126/science.289.5477.284>
- Wald, A. (1949). Statistical decision functions. *The Annals of Mathematical Statistics*, 20(2), 165–205. <https://doi.org/10.1214/aoms/1177730030>
- Wessel, J., Iyer, G., Wild, T., Ou, Y., McJeon, H., & Lamontagne, J. (2024a). Large ensemble exploration of global energy transitions under national emissions pledges. *Earth's Future*, 12(10), e2024EF004754. <https://doi.org/10.1029/2024ef004754>
- Wessel, J., Lamontagne, J., Iyer, G., Wild, T., Ou, Y., & McJeon, H. (2024b). Large ensemble exploration of global energy transitions under national emissions pledges (processed data, figure code, and workflow scripts) [Dataset]. *Zenodo*. <https://doi.org/10.5281/zenodo.12745749>
- Weyant, J. (2017). Some contributions of integrated assessment models of global climate change. *Review of Environmental Economics and Policy*, 11(1), 115–137. <https://doi.org/10.1093/reep/rew018>
- Wild, T. B., Khan, Z., Zhao, M., Suriano, M., Bereslawski, J. L., Roberts, P., et al. (2021). The implications of global change for the co-evolution of Argentina's integrated energy-water-land systems. *Earth's Future*, 9(8), e2020EF001970. <https://doi.org/10.1029/2020ef001970>
- Wise, M., Calvin, K., Kyle, P., Luckow, P., & Edmonds, J. (2014). Economic and physical modeling of land use in gcam 3.0 and an application to agricultural productivity, land, and terrestrial carbon. *Climate Change Economics*, 5(2), 1450003. <https://doi.org/10.1142/s2010007814500031>
- Woodard, D. L., Snyder, A., Lamontagne, J. R., Tebaldi, C., Morris, J., Calvin, K. V., et al. (2023). Scenario discovery analysis of drivers of solar and wind energy transitions through 2050. *Earth's Future*, 11(8), e2022EF003442. <https://doi.org/10.1029/2022ef003442>